# Kiernan Preve

San Luis, Argentina · EU citizenship (Italian) · English C1 (TOEFL 106)

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## Professional Summary

Physicist and maker who builds the whole chain from sensor to insight: low-cost hardware, computer vision, statistics and modeling in Python and R. Finishing a Physics degree at UNSL (expected December 2026) after an experimental thesis in magnetobiology applied to AgTech. Outside the lab I produce music, host a science podcast and publish fiction.

## Key Skills

- **Scientific computing:** Python (pandas, NumPy, SciPy, OpenCV), R, SQL, Julia, statistics, physical modeling
- **Computer vision:** automated counting and measurement from images (OpenCV)
- **Hardware and making:** Arduino (C/C++), DAQ, sensors, open-source hardware, CAD and 3D design
- **Ways of working:** Git/GitHub, Linux, Agile/Scrum, bilingual technical writing

## Experience

**Undergraduate Research Assistant (Thesis)**, Magnetobiology Lab (MBLab), UNSL · 2025–Present

- Ran experimental campaigns on how weak static magnetic fields affect germination vigor and early growth in maize seeds; in-person experiments completed September 2026.
- Built a Python/OpenCV pipeline for automated seed counting and radicle length measurement, replacing manual scoring.
- Analyzed germination data and modeled growth curves in Python, with Holm–Bonferroni correction.

**Scrum Master (Internship)**, MakisanTech · 2024

- Facilitated Agile workflows for a 5-person hardware/software team delivering a web-based appointment booking app.

**Instrumentation Developer**, LibreLab, UNSL · 2021–2022

- Prototyped low-cost sensors and scientific instruments on open-source hardware, including an Arduino-based IoT rain gauge for local hydrological data.
- Integrated microcontrollers with visualization interfaces for real-time monitoring of physical variables.

## Creative Work and Science Communication

- **Music producer:** arrangement, mixing and mastering (Pro Tools, Studio One, Ableton, Cubase); original work and covers on YouTube.
- **Podcast host, *La curiosidad nos va a salvar*:** physics and interdisciplinary research for general audiences.
- **Author:** novel *Dentro del Origen* (ISBN 978-9878242514); second novel *Andra* in progress.
- **NASA Space Apps Challenge:** wrote and narrated a solar-flare outreach story for young audiences.

## Education

- **Licenciatura en Física (5-year Physics degree)**, Universidad Nacional de San Luis · 2018–present · expected graduation December 2026
- **Tecnicatura Universitaria en Producción Musical (Music Production)**, Universidad Nacional de San Luis · 2013–2017

## Languages

Spanish (native) · English (C1, TOEFL 106) · German (A1) · French (A1)